

TECHNOLOGICAL UNIVERSITY DUBLIN
Grangegorman

TU256 - Postgraduate Certificate in Data Science
TU059/TU060 - MSc in Computer Science
Year 1/2

Quiz

SPEC 9270 Machine Learning

Part 1 – Multiple Choice Questions

1) (2 Marks) What does the "k" in kNN algorithm represent?

- A. The number of clusters in the dataset
- B. The number of features considered for each prediction
- C. The number of nearest neighbours considered for voting
- D. The number of categories or classes in the dataset

Solution: C

2) (1 Marks) Cross-validation has no advantage over holdout validation when an abundance of data is available.

- True
- False

Answer: False. Cross-validation has no advantage over holdout validation when abundance of data is available? Cross-validation allows more efficient utilization of data during evaluation since data points become train/test data at least once across different folds.

3) (2 Marks) Which of the following is a disadvantage of using the kNN algorithm?

- A. It requires a predetermined number of clusters.
- B. It is highly sensitive to outliers.
- C. It cannot handle categorical variables.
- D. It explicitly computes decision boundaries.

Solution: B

4) (2 Marks) In a decision tree, what criterion often decides the best split among the features?

- A. Accuracy
- B. Gini Entropy
- C. Information Gain
- D. Cosine similarity

Solution: C

5) (2 Marks) Which of the following is true about decision tree classifiers?

- A. They cannot handle numerical data.
- B. They are prone to overfitting, especially with very deep trees.
- C. They require feature scaling to perform well.
- D. They are based on the principle of entropy minimization in clustering.

Solution: B

6) (2 Marks) What assumption do Naive Bayes classifiers make about the features?

- A. All features are dependent on each other.
- B. Features have a uniform distribution.
- C. Features are mutually independent given the class label.
- D. All features are equally important.

Solution: C

7) (2 Marks) Which of the following is a typical application for Naive Bayes classifiers?

- A. Image recognition
- B. Real-time prediction
- C. Spam email detection
- D. Anomaly detection in network traffic

Solution: C

8) (2 Marks) Which metric is commonly used for evaluating models on imbalanced datasets?

- A. Accuracy
- B. F1 score
- C. Recall
- D. Precision

Solution: B

9) (2 Marks) What does the ROC curve represent?

- A. The trade-off between precision and recall
- B. The relationship between cluster purity and the number of clusters
- C. The trade-off between true positive rate and false positive rate
- D. The relationship between model complexity and training error

Solution: C

- 10) (2 Marks) Cross-validation is a technique used for:
- A. Reducing the dimensionality of the feature space
 - B. Estimating the performance of a model on unseen data
 - C. Increasing the speed of the training process
 - D. Decreasing the likelihood of feature selection bias

Solution: B

- 11) (2 Marks) Select all causes of overfitting in machine learning predictive models.

- Model is too simple,
- Labelling errors in training data
- Lack of training data.

Answer: Labelling errors in training data and Lack of training data. The simplicity of the model often leads to underfitting.

- 12) (2 Marks) As a lazy learning strategy, select all of the following that are true for k-Nearest neighbour:

- The classifier keeps all training examples for later use,
- The classifier uses all training examples in training
- Little work is done offline in training.

Answer: A lazy learner keeps all training examples for later use, doing as little work as possible offline. So the first and the third options apply.

- 13) (1 Marks) Suppose a classifier is trained on 150 instances using both the hold-out and cross-validation methods! The first method gives you an accuracy of 0.934, and the second one gives an accuracy of 0.895! Which method is more reliable, and would you trust the most?

- A. Cross-validation method,
- B. The hold-out method

Answer: Under A). The cross-validation method is more reliable since it takes more dataset slices on multiple repetitions and averages across them as a test set.

- 14) (2 Marks) What is the goal of gradient descent in machine learning?

- A. To maximize the likelihood function
- B. To minimize the cost function
- C. To maximize the margin between classes
- D. To minimize the error rate

Answer: B) To minimize the cost function

15) (2 Marks) In linear regression, what does the term "slope" represent?

- A. The intercept of the regression line
- B. The measure of spread of data points
- C. The change in the dependent variable for a unit change in the independent variable
- D. The degree of correlation between variables

Answer: C) The change in the dependent variable for a unit change in the independent variable

16) (2 Marks) What is the main objective of a support vector machine (SVM) classifier?

- A. To minimize classification error
- B. To maximize the margin between classes
- C. To minimize the number of support vectors
- D. To maximize the distance between data points

Answer: B) To maximize the margin between classes

17) (2 Marks) In support vector machines, what are support vectors?

- A. Data points closest to the decision boundary
- B. Outliers that are removed during training
- C. Data points with the highest weights in the model
- D. Data points with the lowest weights in the model

Answer: A) Data points closest to the decision boundary

18) (2 Marks) Which kernel function is commonly used in support vector machines for handling non-linearly separable data?

- A. Linear kernel
- B. Polynomial kernel
- C. Radial Basis Function (RBF) kernel
- D. Sigmoid kernel

Answer: C) Radial Basis Function (RBF) kernel

19) (2 Marks) In the context of artificial neural networks (ANNs), what does the term "backpropagation" refer to?

- A. The process of updating the weights of the network by propagating errors backward from the output layer to the input layer
- B. The activation function used to introduce non-linearity in the network
- C. The process of randomly initializing the weights and biases of the network
- D. The method of adding multiple layers to the network to increase its depth

Answer: A) The process of updating the weights of the network by propagating errors backward from the output layer to the input layer

20) (4 Marks) Consider you have been assigned to build a simple feedforward neural network to classify emails as either spam or not spam. Each email is represented by a bag-of-words feature vector, where each element corresponds to the frequency of a particular word in the email. Your dataset consists of 10,000 emails, and you have decided to use a single hidden layer neural network for this task. Given this scenario, which of the following choices is the most appropriate consideration when designing the architecture of your neural network?

- A. Using a sigmoid activation function in the output layer to ensure probabilistic outputs between 0 and 1.
- B. Selecting the number of neurons in the hidden layer based on the complexity of the classification task and the size of the feature space.
- C. Initializing all weights and biases to zero to ensure symmetry and avoid vanishing gradients during training.
- D. Using a linear activation function in the hidden layer to enable the network to learn complex non-linear patterns in the data.

Answer:

B) Selecting the number of neurons in the hidden layer based on the complexity of the classification task and the size of the feature space.

Explanation:

When designing a feedforward neural network, the number of neurons in the hidden layer(s) is an important consideration. Choosing the appropriate number of neurons involves balancing model complexity with the capacity to learn from the data. In this case, the size of the feature space (bag-of-words representation) and the complexity of the classification task (spam detection) should guide the selection of the number of neurons in the hidden layer. Considering the priorities given in choices A, C, and D are not the primary considerations when designing the architecture of the neural network for this specific task.

Part 2 – Small Exercises

Answer two of the following questions:

- 1) (10 Marks). Find the distance between the two items in the table given below (using all descriptive features – use Euclidean for continuous and overlap for categorical features), and give your answer correct to 1 decimal place.

ID	Speed	Agility	Strength	Age	Draft	Nationality
1	13.0	10.5	2.1	19	Yes	Irish
2	11.0	5.3	7.1	18	No	Polish

$$\text{Sqrt}((13.0-11.0)^2+(10.5-5.3)^2+(2.1-7.1)^2+(19-18)^2) + (1-0) + (1-0) = 9.552483035399682 \approx 9.6$$

Answer: 9.6

Choose one of the questions:

Consider the following training set consisting of 13 examples and three features.

Weather	Burglar	Cat	Alarm
Rainy	False	True	True
Still	False	False	False
Stormy	True	True	True
Still	False	False	False
Stormy	False	False	False
Still	True	False	True
Rainy	False	False	False
Stormy	True	True	True
Rainy	False	False	False
Stormy	False	False	False
Stormy	True	True	False
Rainy	True	True	True
Stormy	True	True	False

Suppose a Naïve Bayes classifier is trained with these instances.

- 2) (10 Marks). Calculate the probability of Alarm=false for the following query using Naïve bayes to 2 decimal places:

- $P(\text{Weather}=\text{Rainy}, \text{Burglar}=\text{False}, \text{Cat}=\text{False})$
- $\text{Answer} = (2/8) * (6/8) * (6/8) * (8/13) = 0.086538461538462 \approx 0.09$